

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/2024 Semester 1 MH1100 Calculus I

Tutorial 8, Week 9

Your tutor will aim to discuss: Problem 1, 4, 5, 6, 8, 12 and 15

Problem 1 Find dy/dx by implicit differentiation

- (a) $\tan(x/y) = x + y$.
(b) $y \sin(x^2) = x \sin(y^2)$

[Solution:] (a) Treat y as a function of x , and differentiate the equation implicitly with respect to x , we have

$$\sec^2(x/y) \frac{y - xy'}{y^2} = 1 + y'$$

$$y' = \frac{y \sec^2(x/y) - y^2}{y^2 + x \sec^2(x/y)}$$

(b) Treat y as a function of x , and differentiate the equation implicitly with respect to x , we have

$$y' \sin(x^2) + y \sin(x^2) 2x = \sin(y^2) + x \cos(y^2) 2yy'$$

$$y'(\sin(x^2) - 2xy \cos(y^2)) = \sin(y^2) - 2xy \sin(x^2)$$

$$y' = \frac{\sin(y^2) - 2xy \sin(x^2)}{\sin(x^2) - 2xy \cos(y^2)}$$

Problem 2 If $f(x) + x^2(f(x))^3 = 10$ and $f(1) = 2$, find $f'(1)$

[Solution:] We differentiate the equation implicitly with respect to x , and have

$$f'(x) + 2xf^3(x) + x^2(3f^2(x)f'(x)) = 0$$

$$f'(x) = \frac{-2xf^3(x)}{1 + 3x^2f^2(x)}$$

so we have $f'(1) = \frac{-2 \cdot 1 \cdot f^3(1)}{1 + 3 \cdot 1 \cdot f^2(1)} = -\frac{16}{13}$

Problem 3 Find y'' by implicit differentiation

- (a) $x^2 + 4y^2 = 4$;
(b) $\sin y + \cos x = 1$;

[Solution:] (a) We differentiate the equation implicitly with respect to x , and have

$$2x + 4(2yy') = 0$$

$$y' = -\frac{x}{4y}$$

$$y'' = -\frac{1}{4} \frac{y - xy'}{y^2}$$

Bring the equation of y' into the above equation,

$$y'' = -\frac{1}{4} \frac{y - x(-\frac{x}{4y})}{y^2}$$

$$y'' = -\frac{1}{4} \frac{4y^2 + x^2}{y^2(4y)}$$

Note that $x^2 + 4y^2 = 4$, we have

$$y'' = -\frac{1}{4y^3}$$

(b) We differentiate the equation implicitly with respect to x , and have,

$$\cos(y)y' - \sin x = 0$$

$$y' = \frac{\sin x}{\cos y}$$

$$y'' = \frac{\cos x \cos y - \sin x(-\sin(y)y')}{(\cos y)^2}$$

$$y'' = \frac{\cos x \cos y + y' \sin x \sin y}{(\cos y)^2}$$

Bring the equation of y' into the above equation,

$$y'' = \frac{\cos x \cos y + \frac{\sin x}{\cos y} \sin x \sin y}{(\cos y)^2}$$

$$y'' = \frac{\cos x (\cos y)^2 + (\sin x)^2 \sin y}{(\cos y)^3}$$

Problem 4 Two curves are orthogonal if their tangent lines are perpendicular at each point of intersection. Show that the given curves below are orthogonal of each other.

(a) Curve $y = cx^2$ and curve $x^2 + 2y^2 = k$

(b) Curve $x^2 + y^2 = c$ and curve $kx + y = 0$

(Here number c and k are constant value)

[Solution:] (a) We can calculate their derivatives,

$$y' = 2cx$$

and

$$2x + 2(2yy') = 0$$

$$y' = -\frac{x}{2y}$$

If two curves intersect at (x_0, y_0) , then we have $y_0 = cx_0^2$. The multiplication of tangent values of the two curvatures at (x_0, y_0) is,

$$(2cx_0)(-\frac{x_0}{2y_0}) = -\frac{cx_0^2}{y_0} = -1$$

This means that their tangent lines are perpendicular at any intersection points, thus two curves are orthogonal.

(b) We can calculate their derivatives,

$$2x + 2yy' = 0$$

$$y' = -\frac{x}{y}$$

and

$$y' = -k$$

If two curves intersect at (x_0, y_0) , then we have $kx_0 + y_0 = 0$. The multiplication of tangent values of the two curvatures at (x_0, y_0) is,

$$\left(-\frac{x_0}{y_0}\right)(-k) = \frac{kx_0}{y_0} = -1$$

This means that their tangent lines are perpendicular at any intersection points, thus two curves are orthogonal.

Problem 5 Find all points on the curve $x^2y^2 + xy = 2$ where the slope of the tangent line is -1 .

[Solution:] We differentiate the equation implicitly with respect to x , and have

$$2xy^2 + x^2(2yy') + y + xy' = 0$$

$$y' = -\frac{2xy^2 + y}{x + 2x^2y}$$

If the slope of the tangent line at point (x_0, y_0) is -1 , we have

$$-1 = -\frac{2x_0y_0^2 + y_0}{x_0 + 2x_0^2y_0}$$

$$2x_0y_0^2 + y_0 = x_0 + 2x_0^2y_0$$

$$(2x_0y_0 + 1)(y_0 - x_0) = 0$$

Therefore, the potential solution is $x_0y_0 = -1/2$ or $x_0 = y_0$. Since $x_0y_0 = -1/2$ does not satisfy $x_0^2y_0^2 + x_0y_0 = 2$, we can only have $x_0 = y_0$, further bring it to the original equation

$$x_0^4 + x_0^2 = 2$$

$$(x_0^2 - 1)(x_0^2 + 2) = 0$$

thus we have $x_0 = \pm 1$, and the points are $(1, 1)$ and $(-1, -1)$.

Problem 6 Two cars start moving from the same point. One travels south at 30km/h and the other travels west at 72km/h . At what rate is the distance between the cars increasing two hour later?

[Solution:] We can draw a graph as follows,

The distance between two car is defined as y , which satisfies,

$$y^2 = c^2 + b^2$$

we differentiate both sides with respect to time t , and have

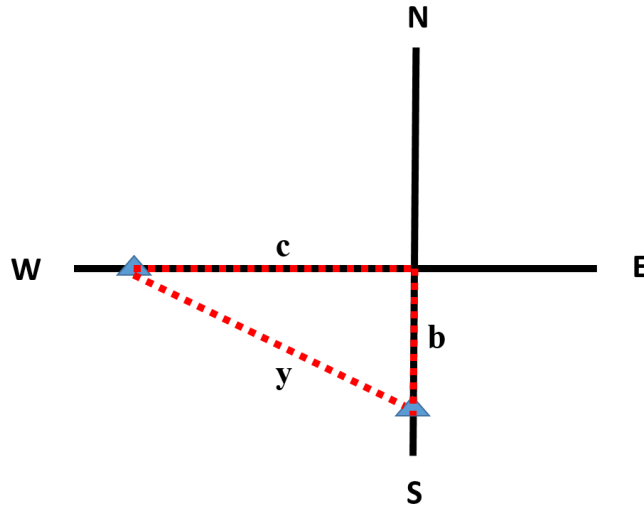
$$2y\frac{dy}{dt} = 2c\frac{dc}{dt} + 2b\frac{db}{dt}$$

$$\frac{dy}{dt} = \frac{c}{y}\frac{dc}{dt} + \frac{b}{y}\frac{db}{dt}$$

From the conditions, we have $\frac{db}{dt} = 30\text{ km/h}$ and $\frac{dc}{dt} = 72\text{ km/h}$. After two hours, $b = 30 \cdot 2 = 60\text{ km}$ and $c = 72 \cdot 2 = 144\text{ km}$. And $y = \sqrt{60^2 + 144^2} = 156\text{ km}$.

Therefore, we can evaluate the rate of distance increasing between two cars,

$$\frac{dy}{dt} = \frac{144}{156} \cdot 72 + \frac{60}{156} \cdot 30 = 78\text{km/h}$$



Problem 7 The height (in meters) of a projectile shot vertically upward from a point 2 m above ground level with an initial velocity of 24.5 m/s is $h = 2 + 24.5t - 4.9t^2$ after t seconds.

(a) When does the projectile reach its maximum height?

(b) What is maximum height?

[Solution:] (a) The velocity is

$$v(t) = \frac{dh}{dt} = 24.5 - 9.8t$$

The projectile reaches its maximum height when $v(t) = 0$ m/s,

$$t = 2.5s$$

(b) The maximum height is

$$h(t = 2.5) = 2 + 24.5 \cdot 2.5 - 4.9(2.5)^2 = 32.625m$$

Problem 8 The edge of a cube was found to be 10 cm with a possible error in measurement of 0.1 cm. Use differentials to estimate the maximum possible error of

(a) the volume of the cube;

(b) the surface area of the cube.

[Solution:] If we denote V , S and h as the volume, surface and edge of the cube. We have

$$V = h^3$$

$$S = 6h^2$$

Therefore we have differentials

$$dV = 3h^2dh$$

$$dS = 12h dh$$

and the maximum possible error is

$$dV = 3 \cdot (10)^2 \cdot 0.1 = 30 \text{ cm}^3$$

$$dS = 12 \cdot 10 \cdot 0.1 = 12 \text{ cm}^2$$

Problem 9 Find the limit

$$\lim_{x \rightarrow 0} \frac{\sin 3x \cdot \sin 5x}{x^2}$$

[Solution:] Note that

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin 3x \cdot \sin 5x}{x^2} &= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot \frac{\sin 5x}{5x} \cdot \frac{15x^2}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \cdot \lim_{x \rightarrow 0} \frac{15x^2}{x^2} \\ &= 15 \end{aligned}$$

Therefore we have

$$\lim_{x \rightarrow 0} \frac{\sin 3x \cdot \sin 5x}{x^2} = 15$$

Problem 10 If $F(x) = f(3f(4f(x)))$, where $f(0) = 0$ and $f'(0) = 2$, find $F'(0)$.

[Solution:] Note that

$$F'(x) = f'(3f(4f(x))) \cdot 3f'(4f(x)) \cdot 4f'(x)$$

Therefore we have

$$\begin{aligned} F'(0) &= f'(3f(4f(0))) \cdot 3f'(4f(0)) \cdot 4f'(0) \\ &= f'(3f(4 \cdot 0)) \cdot 3f'(4 \cdot 0) \cdot 4f'(0) \\ &= f'(3 \cdot 0) \cdot 3 \cdot 2 \cdot 4 \cdot 2 \\ &= 2 \cdot 3 \cdot 4 \cdot 2 \cdot 2 \\ &= 96 \end{aligned}$$

Problem 11 Write the composite function in the form $f(g(x))$. [Identify the inner function $u = g(x)$ and the outer function $y = f(u)$.] Then find the derivative dy/dx .

$$(a) y = \sqrt{4+3x} \quad (b) y = \tan(\sin x) \quad (c) y = \sin \sqrt{x} \quad (d) y = (1-x^2)^{100}$$

[Solution:]

(a) Define

$$f(x) = \sqrt{x} \quad \text{and} \quad g(x) = 4 + 3x,$$

we can express y as

$$y(x) = \sqrt{4+3x} = (f \circ g)(x) = f(g(x)).$$

Since

$$f'(x) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}} \quad \text{and} \quad g'(x) = 3,$$

we have

$$y'(x) = f'(g(x)) \cdot g'(x) = \frac{1}{2\sqrt{4+3x}} \cdot 3 = \frac{3}{2\sqrt{4+3x}}$$

(b) Let

$$f(x) = \tan x \quad \text{and} \quad g(x) = \sin x.$$

We can express y as

$$y(x) = \tan(\sin x) = (f \circ g)(x) = f(g(x)).$$

Since

$$f'(x) = \sec^2 x \quad \text{and} \quad g'(x) = \cos x,$$

we have

$$y'(x) = f'(g(x)) \cdot g'(x) = \sec^2(\sin x) \cdot \cos x = \sec^2(\sin x) \cdot \cos x.$$

(c) Let

$$f(x) = \sin x \quad \text{and} \quad g(x) = \sqrt{x}.$$

We can express y as

$$y(x) = \sin \sqrt{x} = (f \circ g)(x) = f(g(x)).$$

Since

$$f'(x) = \cos x \quad \text{and} \quad g'(x) = \frac{1}{2\sqrt{x}},$$

we have

$$y'(x) = f'(g(x)) \cdot g'(x) = \cos(\sqrt{x}) \cdot \frac{1}{2\sqrt{x}} = \frac{\cos(\sqrt{x})}{2\sqrt{x}}.$$

(d) Let

$$f(x) = x^{100} \quad \text{and} \quad g(x) = 1 - x^2.$$

We can express y as

$$y(x) = (1 - x^2)^{100} = (f \circ g)(x) = f(g(x)).$$

Since

$$f'(x) = 100x^{99} \quad \text{and} \quad g'(x) = -2x,$$

we have

$$y'(x) = f'(g(x)) \cdot g'(x) = 100(1 - x^2)^{99} \cdot (-2x) = -200x(1 - x^2)^{99}.$$

Problem 12 Find y' and y'' .

$$(a) \ y = \cos(\sin 3\theta) \qquad (b) \ y = \frac{1}{(1 + \tan x)^2} \qquad (c) \ y = \frac{4x}{x + 1}$$

[Solution:]

(a)

$$\begin{aligned}
y' &= -\sin(\sin 3\theta) \cdot (\sin 3\theta)' \\
&= -\sin(\sin 3\theta) \cdot (\cos 3\theta) \cdot (3\theta)' \\
&= -\sin(\sin 3\theta) \cdot (\cos 3\theta) \cdot 3 \\
&= -3\sin(\sin 3\theta) \cos 3\theta. \\
y'' &= -3[\sin(\sin 3\theta)]' \cos 3\theta - 3\sin(\sin 3\theta)(\cos 3\theta)' \\
&= -9\cos(\sin 3\theta) \cos 3\theta \cos 3\theta + 9\sin(\sin 3\theta) \sin 3\theta \\
&= -9\cos(\sin 3\theta) \cos^2 3\theta + 9\sin(\sin 3\theta) \sin 3\theta.
\end{aligned}$$

(b)

$$\begin{aligned}
y' &= -2(1 + \tan x)^{-3} \cdot (1 + \tan x)' \\
&= -2(1 + \tan x)^{-3} \cdot \sec^2 x \\
&= \frac{-2\sec^2 x}{(1 + \tan x)^3}. \\
y'' &= \frac{(1 + \tan x)^3(-2\sec^2 x)' - (-2\sec^2 x) [(1 + \tan x)^3]'}{(1 + \tan x)^6} \\
&= \frac{(1 + \tan x)^3(-4\sec^2 x \tan x) - (-2\sec^2 x) [3(1 + \tan x)^2(1 + \tan x)']}{(1 + \tan x)^6} \\
&= \frac{(1 + \tan x)^3(-4\sec^2 x \tan x) - (-2\sec^4 x) [3(1 + \tan x)^2]}{(1 + \tan x)^6} \\
&= \frac{-4(1 + \tan x)^3 \sec^2 x \tan x + 6\sec^4 x(1 + \tan x)^2}{(1 + \tan x)^6} \\
&= \frac{-4\sec^2 x \tan x}{(1 + \tan x)^3} + \frac{6\sec^4 x}{(1 + \tan x)^4}.
\end{aligned}$$

(c) We can first rewrite y as

$$y = \frac{4x}{x+1} = \frac{4x+4-4}{x+1} = 4 - 4(x+1)^{-1}.$$

Hence,

$$\begin{aligned}
y'(x) &= 4(x+1)^{-2} \cdot (x+1)' \\
&= \frac{4}{(x+1)^2}
\end{aligned}$$

and

$$\begin{aligned}
y''(x) &= 4(-2)(x+1)^{-3} \cdot (x+1)' \\
&= -\frac{8}{(x+1)^3}.
\end{aligned}$$

Problem 13 Suppose f is differentiable on $\mathbb{R}$ and α is a real number. Let $F(x) = f(x^\alpha)$ and $G(x) = [f(x)]^\alpha$. Find expressions for (a) $F'(x)$ and (b) $G'(x)$.

[Solution:] We use the chain rule to get the two derivatives.

(a)

$$\begin{aligned}
F'(x) &= f'(x^\alpha) \cdot (x^\alpha)' \\
&= f'(x^\alpha) \cdot (\alpha x^{\alpha-1}) \\
&= \alpha x^{\alpha-1} f'(x^\alpha).
\end{aligned}$$

(b)

$$\begin{aligned}
G'(x) &= \alpha [f(x)]^{\alpha-1} \cdot f'(x) \\
&= \alpha [f(x)]^{\alpha-1} f'(x).
\end{aligned}$$

Problem 14 If g is a twice differentiable function and $f(x) = xg(x^2)$, find f'' in terms of g , g' , and g'' .

[Solution:] The first derivative of f is

$$\begin{aligned}
f'(x) &= x'g(x^2) + x[g(x^2)]' \\
&= g(x^2) + xg'(x^2) \cdot (x^2)' \\
&= g(x^2) + 2x^2g'(x^2).
\end{aligned}$$

The second derivative is therefore

$$\begin{aligned}
f''(x) &= g'(x^2) \cdot (x^2)' + (2x^2)'g'(x^2) + 2x^2g''(x^2) \cdot (x^2)' \\
&= g'(x^2) \cdot (2x) + (4x)g'(x^2) + 2x^2g''(x^2) \cdot (2x) \\
&= 2xg'(x^2) + 4xg'(x^2) + 4x^3g''(x^2) \\
&= 6xg'(x^2) + 4x^3g''(x^2).
\end{aligned}$$

So,

$$f''(x) = 6xg'(x^2) + 4x^3g''(x^2).$$

Problem 15 If $F(x) = f(xf(xf(x)))$, where $f(1) = 2$, $f(2) = 3$, $f'(1) = 4$, $f'(2) = 5$, and $f'(3) = 6$, find $F'(1)$.

[Solution:] Using the chain rule, the derivative of $F(x)$ is

$$\begin{aligned}
F'(x) &= f'(xf(xf(x))) \cdot (xf(xf(x)))' \\
&= f'(xf(xf(x))) \cdot [x'f(xf(x)) + xf'(xf(x)) \cdot (xf(x))'] \\
&= f'(xf(xf(x))) \cdot [f(xf(x)) + xf'(xf(x)) \cdot (f(x) + xf'(x))] \\
&= f'(xf(xf(x))) [f(xf(x)) + xf'(xf(x))f(x) + x^2f'(xf(x))f'(x)].
\end{aligned}$$

Given that $f(1) = 2$, $f(2) = 3$, $f'(1) = 4$, $f'(2) = 5$, and $f'(3) = 6$, we have

$$\begin{aligned}
F'(1) &= f'(f(f(1))) [f(f(1)) + f'(f(1))f(1) + f'(f(1))f'(1)] \\
&= f'(f(2)) [f(2) + f'(2)2 + f'(2)4] \\
&= f'(3) [3 + 5 \times 2 + 5 \times 4] \\
&= 6 \times 33 \\
&= 198.
\end{aligned}$$